



CollaborAITE

AM Peer Review

Prepared by The Slice Is Right

Collaborators: Bryan S, Cooper S, Ethan W, Greg M, Israk A

Client: Terry Yoo

COS 497 - Computer Science Capstone 2

April 4th, 2026

Version 1.0.0

1. Structural & Formatting Issues

Location	Issue	Recommendation
Section 2 & 3	Placeholder text [1-2 sentences introducing the major section] appears at the start of both sections.	Replace with actual introductory sentences. For Section 2: "This section describes the high-level architecture, external dependencies, and hardware/software prerequisites for running CollaborAITE." For Section 3: "This section provides step-by-step procedures for all administrative tasks, from initial installation to daily operations and user support."
Page 5, Table 4	The table mixes client-side requirements (browser, cookies) with backend dependencies (Socket.IO, Firebase Admin SDK).	Split into two tables: Table 4a: Client-Side (End User) Requirements and Table 4b: Backend Software Dependencies (moving Socket.IO, Firebase Admin SDK, Anthropic/OpenAI SDKs, etc. there).
Section 3.1, Step 3	Environment variable listing uses inconsistent formatting (e.g., VITE_API_URL with no example, AI_DEFAULT_PROVIDER \(\equiv\) claude with a strange symbol).	Clean up formatting. Use consistent KEY=value examples. Replace \(\equiv\) with = or (default: claude).

Section 3.2 & 3.3	Several sub-headings are repeated verbatim (e.g., "Production Deployment:" appears twice; "Updating AI Provider API Keys:" appears twice under different sections).	Remove duplicate headings. Keep one instance of each under the most relevant section (e.g., "Production Deployment" under 3.1 Installation, "Updating AI Provider API Keys" under 3.2 Routine Tasks).
Throughout	Numbered lists often start with 0 instead of 1. (e.g., Step 4, Step 7, troubleshooting items).	Replace leading zeros with proper numbering or bullet points.

2. Content Gaps & Missing Information

Location	Issue	Recommendation
Section 2.2	The sentence [Add a few sentences explaining the tables] is left as a placeholder.	Add a brief explanation, e.g., "The following tables list the minimum and recommended infrastructure, development machine specifications, key software dependencies, and client-side requirements for using CollaborAITE."
Section 3.1, Step 6	The seeding script command <code>npm tsx scripts/seed-dev-group.ts</code> is incorrect. <code>npm</code> does not directly run <code>tsx</code> .	Change to <code>npx tsx scripts/seed-dev-group.ts</code> or, if <code>tsx</code> is a defined <code>npm</code> script, clarify as <code>npm run tsx -- scripts/seed-dev-group.ts</code> .
Section 3.2 – Monitoring AI API Usage	Mentions "in-application analytics" accessible to group owners/admins, but no guidance is given on where to find these analytics in the UI.	Add a sentence: "Group owners and admins can access these analytics by navigating to Group Settings > Analytics tab within the CollaborAITE web interface."

Section 3.3 – Database Backups	The command uses <code>gs://your-backup-bucket</code> but doesn't instruct the admin to create the bucket first or set lifecycle policies.	Add prerequisite: <code>*"Before running the export, create a Cloud Storage bucket (e.g., <code>gs://collaboraite-backups</code>) and optionally configure a lifecycle rule to auto-delete backups older than 30 days."*</code>
Section 3.3 – Cleaning Up Stale Data	Mentions manual purging of expired invite tokens but provides no script or console path.	Provide a concrete example: <code>"To purge tokens older than 7 days, run the following in Firebase Console > Firestore > inviteTokens collection: delete documents where expiresAt < current time. Alternatively, use the provided script at scripts/purge-expired-tokens.js."</code>
Section 3.4 – User Support	No mention of how administrators should handle password resets or account recovery, even though authentication is Google OAuth-only.	Add a note: <code>"Because CollaborAITE uses Google OAuth 2.0 exclusively, password resets and account recovery are managed entirely by Google. Administrators cannot reset user passwords directly; users must use Google's account recovery flow."</code>

Section 4.2 – Known Bug #7

Email invitation is a "console stub." No workaround is given beyond using shareable invite links.

Expand workaround: "Group owners can generate shareable invite links from the Group Members panel by clicking 'Invite via Link'. These links are valid for 7 days by default and can be copied and distributed manually (e.g., via institutional email or LMS)."

Appendix A

The client signature line shows "Dr. Greg Nelson" but the document front page and references use "Dr. Gregory Nelson."

Standardize the client's name throughout the document.

3. Technical Accuracy & Consistency

Location	Issue	Recommendation
Section 3.1, Step 3	VITE_USE_REAL_AUTH=false is described as using Firebase emulators, but emulators require additional configuration (e.g., GCLOUD_PROJECT).	Add a note: "When using emulators, also set GCLOUD_PROJECT=your-project-id and ensure the Firebase Emulator Suite is running with Auth emulator enabled."
Section 3.2 – Reviewing Logs	States "Sentry integration is planned but not yet configured." This is stale planning language.	Either confirm Sentry is now configured (update the doc) or remove the forward-looking statement and simply state current logging limitations.
Section 4.1 – Toast error "TypeError"	States "the error claim from the sibling team's CIR was incorrect - no TypeError actually occurs." This is defensive and unprofessional in an admin manual.	Remove the blame language. Simply state: "This error does not occur in practice. Both destructuring patterns function correctly. The issue has been marked as resolved in the codebase."

Section 4.2 – Known Bug #2	In-memory rate-limit store "does not evict expired entries." Contradicts Section 3.3 which says "the in-memory rate-limit store on the backend does not evict expired entries automatically (known limitation #24 from CIR)."	Merge these references. In Section 3.3, simply cross-reference: <i>"See Section 4.2, Known Bug #2 for details on the rate-limit store limitation."</i> Do not repeat the full description.
Section 4.2 – Known Bug #10	require() inside streaming callback – described as having "no functional impact."	Add a note on potential performance: "While not a bug, this can cause a slight delay on the first streamed response. Fix scheduled to hoist to module scope."

4. Security & Compliance Gaps

Location	Issue	Recommendation
Section 2.1	Mentions "ensuring FERPA compliance through proper data isolation and pseudonymization" but no specific procedures are given in Section 3.	Add a subsection under 3.3 or 3.4 titled FERPA Compliance Checklist that includes: verifying data deletion upon account removal, audit logging export steps, and ensuring pseudonymized export formats.
Section 3.2 – Updating AI Provider API Keys	Tells admins to update keys in .env.local, GitHub Secrets, and Cloud Run environment variables. This is error-prone.	Recommend a single source of truth: "For production, always update API keys via Cloud Run environment variables or GCP Secret Manager. Use GitHub Secrets only for CI/CD builds, not for runtime configuration. Local .env files are for development only."
Section 4.2 – Known Bug #1 (CORS)	"Will fix before production release." The document is dated March 26, 2026. If this is a post-release manual, the fix should already be applied.	Update status to either Fixed in v1.0.1 (with configuration instructions) or, if still unfixed, add a hard warning: "DO NOT deploy to production without restricting CORS. See Appendix C for production CORS configuration."

5. Usability & Clarity Improvements

Location	Issue	Recommendation
Section 1.1	Assumes familiarity with "command-line tools, Node.js, Firebase/Google Cloud Platform, and basic web application deployment concepts." No "quick start" for less-experienced admins.	Add a Prerequisites Checklist sidebar: <code>"Before installing, ensure you have: Node.js v22 LTS, npm v10+, a Google Cloud Project with billing enabled, Firebase project linked to that GCP project, and API keys for at least one AI provider."</code>
Section 3.1, Step 5	Requires two terminals but doesn't explain how to verify both services are healthy together.	Add a verification step: "After both terminals are running, open <code>http://localhost:5173</code> . The backend health check should be automatically pinged on login. You can also manually verify by visiting <code>http://localhost:3001/health</code> ."
Section 3.4, Table 5	The "@AI mentions not working" row suggests checking rate limits but doesn't tell the admin how to view or adjust the limit.	Add: "To adjust rate limits, set the environment variable <code>AI_RATE_LIMIT_PER_USER</code> (default 20) in Cloud Run service configuration and redeploy."

Section 4.1 – Build failures

Recommends `rm -rf node_modules` but does not mention clearing Nx cache, which often causes persistent monorepo issues.

Add: "If issues persist after reinstalling, also clear Nx cache with `nx reset` before rebuilding."

6. Minor Copyediting & Polish

Location	Issue	Correction
Page 2 (ToC)	References section shows 1.1 Purpose of This Document then immediately References without a sub-number.	Re-number References as 1.2 References for consistency.
Page 5, Table 3	Node.js requirement shows V22 + LTS then V22 LTS.	Change to V22 LTS for both Minimum and Recommended.
Page 7, Step 7	Bullet points start with 0 and have inconsistent indentation.	Replace with standard - bullets or 1., 2., 3.
Page 9, Table 5	HTML entities like > appear instead of >.	Replace > with > in the "User cannot see a group" row.
Appendix B	All five team members claim exactly 20% contribution. This is mathematically convenient but unlikely.	Either change to a range (e.g., "15-25%") or add a note: "Effort was evenly distributed across all team members."